Supplement: methods, sensitivities and correction record

Draft for review, 9 November 2026. Barry W. Brook*[1], Claude Opus 5.5 (Anthropic) and GPT-6 Codex (OpenAI)[2]

Draft for review; release and submission are Barry W. Brook's decision.

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This supplement gives the machinery behind the article. Every quoted number is bound to the claims ledger, `manuscript/claims.csv`. The interpretation specification, `manuscript/estimands.md`, states what each analysis bounds and which phrasings are not allowed. Section and decision labels (A04, A10, D018 and so on) refer to the project record in `docs/` and `coordination/decisions.md`.

S1. Definitions, readings and units

Term	Meaning
Issuance	One forecast, issued when the observation-time rule first confirms that an outburst has ended
Duration	Onset-to-end time of the completed outburst, from X4 run on the 400 days before issue
Quiet gap	End of one outburst to onset of the next; the forecast target
Previous-gap predictor, q	Standardised log of the preceding quiet gap, used only if covered (no visit gap over seven days)
Recent-level predictor, L	Standardised mean of up to six earlier covered log quiet gaps; zero after standardisation if none
Onset bracket	X4's retrospective interval for the next onset; scoring evidence only
Right censoring	The forecast ends at coverage loss, the next issuance, or the block end before an onset is established
Log score	Natural log of the probability the forecast gave to the observed bracket (or to survival, if censored), conditional on survival to issue; units are nats
Paired gain	M3's log score minus M2's for the same issuance
Geometric mean likelihood ratio	exp(mean paired gain); a 1.7% increase for the primary
Working clock scale	The median of the untruncated lognormal clock; ratios of scales are model quantities, not observed mean waits
Partial-identification bounds	The smallest and largest coverage consistent with bracketed or censored outcomes

Times are Julian Dates as returned by the archive, without conversion. Durations and gaps are in days. Log durations are standardised by the frozen development centre and scale.

S2. Methods

S2.1 Observations and extraction

Rows were requested in bounded annual windows, ending strictly before 2018. Responses near the 50,000-row service cap were bisected, and every response was hashed on receipt before parsing. The analysis uses rows in the `Vis.` band with validation flag `V`. The 1960–2017 evaluation window contains 63 validated leaf responses. These hold 500,479 rows after removing 81 duplicate identifiers, of which 249,356 meet the analysis selector.

X4 (`src/dnmem/extract_x4.py`, definition `a04f-x4-v1`) groups observer visits into candidate episodes brighter than visual magnitude 10. For each episode it forms a posterior for the brightening and fading threshold crossings, using a slope mixture over nearby estimates plus a shared 0.05-mag systematic level error. That error stops intervals narrowing as the inverse square root of the number of observers, which caused undercoverage in its predecessor. An episode is confirmed by bright reports from two distinct observers, or from one observer on at least two nights with a margin past the threshold and no contradicting faint report. The observation-time confirmation rule arms only on bright evidence at least 0.5 mag past the threshold, so noise near the threshold in a decline cannot re-arm it.

Three earlier extractors, X1 to X3, failed qualification. X4 passed a synthetic screen at magnitude 10 on the qualified phase-aligned cadence family, and the C1/7 d coverage rule was selected there. The unseen waveform family failed benchmark qualification before X4 recovery was scored; X4 recovery on that family was not tested. Magnitude-9.5 diagnostics failed in later eras. Those failures are retained (A04e–A04i). The evaluation is therefore conditional on the magnitude-10 X4 extraction.

S2.2 Issuance and predictors

Each observation-time end confirmation issues one forecast at its emission time. A two-day reporting delay is a sensitivity. Every emission is kept, including false and repeat emissions. Predictors come from X4 run on the rows in the 400 days up to the emission and on nothing else. A unit test inserts poisoned rows after the emission and checks that no ledger field changes. The current episode is the last prefix episode whose end bracket reaches the emission's bright evidence. A duration of zero or less makes the mark unavailable; nothing is clipped. The previous gap requires the previous episode to begin more than 30 days inside the window.

S2.3 Outcomes and censoring

The outcome is the next onset found by retrospective X4 run up to an explicit cutoff. An onset bracket that straddles issue does not establish survival, and that forecast is excluded under the frozen rule. Here none were excluded. A forecast is scored on the onset interval if the interval is established before the first of these events:

- coverage loss, the first visit gap longer than seven days after issue;
- supersession by the next issuance;
- the end of the block.

Otherwise the forecast is right-censored at the earliest of these events.

S2.4 Models and likelihood

Each model is a lognormal clock for the quiet gap Q after a known end e, with log-location μ and log-standard deviation σ:

- M0: $\mu = \alpha$;
- M2: $\mu = \alpha + \gamma_1 q + \gamma_2 L$;
- M3: $\mu = \alpha + \gamma_1 q + \gamma_2 L + \beta x$, with $\beta \geq 0$.

Here $q = (\log Q_{\text{prev}} - c_q)/s_q$. If ℓ is the mean of up to six earlier covered log quiet gaps, $L = (\ell - c_L)/s_L$; set $L = 0$ if none are available. The centres and scales are frozen development means and standard deviations. Likewise, x is the standardised log duration. At issue age a, an onset bracket $(l, u]$ scores $\log P(l - e < Q \leq u - e) - \log S(a)$. A censoring time c scores $\log S(c - e) - \log S(a)$, where S is the survival function. Forecasts without a covered previous gap use M0 in both models, and forecasts without a known end use a common remaining-wait forecast. Both contribute exactly zero to the contrast.

M2 was fitted on timing-eligible 1910–1959 rows and frozen. The duration offset was then fitted on rows with a valid mark, with M2 held fixed, so that $\beta = 0$ reproduces M2 exactly (D021). Frozen values (C04): $\beta =$ 0.046597 per standard deviation of log duration; previous-gap slope 0.078016; recent-level slope 0.051336; σ = 0.34636. The parameter file's SHA-256 begins `ce4ef8d5`.

The constraint $\beta \geq 0$ and matched routing respond to a measurement coupling. The duration and the following gap share the estimated end time, so end-time error couples them. In earlier synthetic work that coupling pushed fitted null coefficients negative (D020). Codex then showed that a nonnegative bound is a restriction of direction, not a theorem about the sign of the error. An exact construction with constant true duration can yield positive information (D021). Endpoint coupling therefore remains a stated limit. The uniform-end and two-day-delay sensitivities probe it only partly.

S2.5 Primary statistic and inference

The statistic is the mean paired gain over all score-eligible issuances with issue times in JD 2436934.5 to 2458119.5. The standard error is cluster-robust, over consecutive clusters of 6×365.25 days anchored at the window start, with the factor $G/(G-1)$ for G nonempty clusters. Inference uses Student's t with $G - 1$ degrees of freedom. The result counts as confirmed if the one-sided 95% lower bound exceeds zero. The cluster length was chosen before any operating characteristics were seen (D021). There is one confirmatory statistic. No subset, era, variant or interval width may replace it.

S2.6 Operating characteristics

Before the development fit, a finite synthetic screen checked the design (A10d). At an effect sized to CM92's published correlations, 11 of 40 simulated evaluations confirmed, with exact interval 0.146 to 0.439. Pooled no-effect histories gave 3 confirmations in 220. These histories share random streams, so the count is a finite

screen, not an independent error rate or a certified size. The reference injected effect gave a mean gain of 0.0045 nats. Barry accepted the finite screen before the first real fit (D022).

S2.7 Access, execution and repair

Stage (8 November 2026, UTC)	Time
Protocol published (DOI 10.5281/zenodo.24053338)	12:34:47
First attempt began; failed before any ledger or score	12:38:28; 12:39:51
Repaired acquisition began	12:51:18
Candidate and reference ledger concordance passed	13:06:58.627
Primary result written	13:06:58.641

The first attempt failed when the archive's literal `uncert="None"` token for missing uncertainty reached a strict parser. Uncertainty enters no extraction, feature, evidence, routing or score path. The repair accepts that exact token without changing or dropping any row. Synthetic invariance tests, independent review, the full test suite, a new full rehearsal and supported-platform CI passed before continuation (D026). All 26 original forecasting source files, the parameter bytes and the protocol bytes are unchanged. The repair freeze is `5e39aeb` and the execution commit `d7151e0`. No person or agent inspected real scores, outcomes, counts or coverage summaries before the primary file existed.

Two independent ledger and evidence implementations had to agree exactly at both delays before any score. A separate checker reproduced all 2,641 variant score rows, with a maximum discrepancy of order 10^{-13}, and the primary interval. Claude recomputed the interval by a separate route. The primary population and routing are in C03.

S3. Results and sensitivities

S3.1 Primary, eras, class rule and sensitivities

Table S1. Frozen comparisons, mean paired log-score gain (nats per issuance) and 90% cluster interval. Only the first row is confirmatory.

Comparison	Issuances	Gain	Interval
Primary, M3 − M2, 1960–2017	441	0.0165	0.0049 to 0.0281
Segment 1960–1992	261	0.0240	0.0091 to 0.0389
Segment 1993–2017	180	0.0057	−0.0107 to 0.0221
12-day class and within-class rule	441	0.0025	−0.0022 to 0.0072
Uniform-end marks	441	0.0166	lower 0.0044
Previous-gap-only baseline	441	0.0170	lower 0.0054
Two-day issue delay	441	0.0165	lower 0.0049
Joint-valid rows	436	0.0167	lower 0.0049
Three-year clusters	441	as primary	lower 0.0059
Baseline M2 − M0	441	0.1168	0.0663 to 0.1672

The primary interval uses ten nonempty blocks and nine degrees of freedom. Eight have positive means. Leave-one-block-out means span 0.0132 to 0.0194, and 0.5397 of individual issuance gains are positive. Holding the estimate, critical value and blocks fixed, the strict failure boundary is approximately 1.42705 times the reported standard error. A 42% inflation passes and an exact 43% fails. None of these checks validates the cluster dependence assumption.

The frozen P3 placebo equals M2, because its development lag coefficient sat at its bound of zero. The planned memory placebo therefore carries no information.

S3.2 Calibration

M3's nominal 50% central intervals covered between 0.546 and 0.617 of outcomes; its nominal 90% intervals between 0.923 and 0.980. Each range runs from definitely covered to possibly covered, given the extracted brackets and censoring, over all scored issuances. These are realised-sample bounds, not confidence intervals for a population rate, and they show overcoverage. Mid-CDF deciles of resolved onsets are descriptive; no uniformity test is claimed.

S3.3 Evaluation-only offsets

Duration offsets were fitted to the evaluation outcomes with an evaluation-refitted M2 held fixed. The support intervals use a likelihood-ratio cutoff of 3.841, with no dependence or measurement calibration.

Offset	Estimate	Conditional support
Current duration, separate	0.04209	0.01574 to 0.06845
Previous duration, separate	−0.03836	−0.06489 to −0.01184
Current duration, joint	0.03045	−0.00063 to 0.06152
Previous duration, joint	−0.02213	−0.05340 to 0.00913

The joint design is correlated but has full rank, with condition number 1.79. Both joint support intervals include zero. Correlation alone neither explains that nor makes the coefficients nonidentifiable.

S3.4 A slow component can hide behind negative lag signs

Negative lag signs alone cannot exclude a dominant slow driver (A10k). Take independent unit-variance AR(1) components S and A, with persistence 0.95 and −0.9, and set log duration $= 0.175S + 0.350A$ and log gap $= 0.300S + 0.001A$. The lag-one duration correlation is then about −0.53, and the lag coefficient conditional on the preceding gap is negative. Yet S dominates gap variation and the current conditional covariance. This construction is a logical counterexample, not fitted to SS Cyg and not a rival empirical result.

S3.5 Exploratory post-result calculations

Everything in this subsection reuses the evaluation outcomes. Each interval conditions on the fitted adjustment it describes. None supplies a further held-out confirmation.

- **Level shift and recentring (C20).** The best common M2 log-location shift is −0.04679. It reduces the original untruncated clock scale by about 4.57%. The frozen duration term implies a level shift of −0.01108, because the evaluation mean standardised log duration is −0.2379. Shifting M2 by that implied amount accounts for 0.00370 nats (0.00013 to 0.00726). The contrast after recentring the duration at its evaluation mean is 0.01283 (0.00359 to 0.02206). With both models at the best M2 level, the gain is 0.01239 (0.00098 to 0.02381). The two orderings of the split differ by about 0.000012 nats.
- **Class versus continuous duration (C21).** With M2's historical parameters fixed and a free common level, the evaluation likelihood-ratio comparisons are as follows:
  - class versus level only, 14.7;
  - continuous log duration versus level only, 11.7;
  - within-class duration added to class, 0.17.

  The class contrast corresponds to a 14% larger untruncated working-clock scale after long outbursts. In 1910–1959 the class and continuous fits were similar, with a smaller contrast.
- **Shared level for the class rule (C22).** The frozen class-plus-within rule's mean offset is 0.01728. Giving both rules M2's evaluation-fitted level yields a gain of 0.00905 (0.00409 to 0.01401). For 1993–2017 alone it yields 0.00761 (−0.00631 to 0.02153). The uncentred type component cannot shorten short-outburst forecasts, but the full frozen rule does shorten 181 of 245 of them through its within-class term.
- **Era refits (C23).** The class contrast is 0.16323 (statistic 13.504) in 1960–1992 and 0.07828 (2.262) in 1993–2017. These are descriptive; there is no test of change.
- **Alternation (C24).** Across forecast-record pairs, the lag-one correlation of log duration is −0.5233 in the evaluation years and −0.4006 in the development years. A long outburst followed a long one in 0.1495 of pairs and a short one in 0.6640. The pairs include one repeat-flagged record and are not a catalogue of unique consecutive outbursts.

S3.6 Development fit

In 1910–1959 the duration offset was 0.0466 per standard deviation, with likelihood-ratio statistic 5.1935 against M2. This fit is in-sample and overlaps CM92's record, so it is not evidence for the claim.

S3.7 Prior art checked against the source

Claude re-read CM92's Table 4 and its Fig. 5 subscript conventions on the original scan (Cannizzo and Mattei 1992) . For 705 outbursts (1896–1992), the correlations are:

- duration with the following quiescence, 0.223;
- duration with the preceding quiescence, 0.282;
- successive durations, −0.400.

Durations split at 12 days, with 401 shorter and 304 longer. The correlation of duration with the following onset-to-onset cycle is 0.475, against 0.145 with the preceding cycle. For end-to-end cycles the pattern reverses, at 0.093 and 0.521. The famous relation is therefore largely definitional, as CM92 showed.

S3.8 Bounded independent novelty check

An independent search in A10m found no prior preregistered probabilistic held-out recurrence evaluation, or explicit held-out comparison of duration beyond a timing-history baseline, among the studies located and checked. This supports a bounded "to our knowledge" statement, not proof of universal priority. The exact queries and source receipts are in `data/provenance/a10m_novelty/`; the appraisal is `docs/audits/2026-11-09-a10m-novelty-review.md`.

The closest precedent, Grigorievskiy et al. (2014), uses chronological training, validation and test sets, with one test outburst of V363 Lyr. It scores point-onset errors; no preregistration or scored predictive onset distribution is reported. No explicit duration-addition comparison against a common timing baseline is reported. Repeated random fits do not make the evaluated forecast a probability distribution. The independent search included an OpenAlex citing-title screen and primary-source checks of this study and Price et al. (2007). ADS access failed; observer/non-English coverage is incomplete, and some full texts, including Tsuranova et al. (2026), remain inaccessible. Their unverified methods are not treated as exclusions.

S4. Analyses superseded during review

Each entry records what was replaced, who found the problem, and where the record is.

Earlier analysis or statement	Problem and finder	Replacement	Record
Duration predicts the following onset-to-onset cycle	Shared term with the cycle (CM92); confirmed in the A01 audit	Quiet-gap target only	A01 memo
Extractors X1, X2, X3	Failed synthetic qualification (Codex and Claude reviews)	X4, qualified at magnitude 10 only	A04e–A04i
Unrestricted duration coefficient on all routes	End-time coupling and route mixing (Claude, D020)	$\beta \geq 0$, matched routing, local gap level	D020, D021
Joint refit of M3 with M2	Training loss not matched to deployment (Codex, D021)	Mark offsets fitted with M2 fixed	D021
Candidate-authority concordance rule	Allowed mismatch deletion (Codex, D024)	Exact two-implementation concordance or abort	D024
Strict parser for the uncertainty field	Failed on `None` after access (execution)	Exact-token repair, reviewed and rehearsed	D026

Interpretation corrections checked in A10k. Codex's independent audit narrowed several readings from Claude's A10j review before the products were drafted. Claude accepted all of them (D030).

Earlier reading	Corrected scope	Evidence
Global recentering isolates drift-free information	It defines an approximately additive descriptive score contrast; within-period drift remains possible	
All predictive information is in outburst type	A class indicator has greater in-sample likelihood; within-class equivalence was not tested	
The entire frozen class rule cannot shorten short-event forecasts	That restriction concerns its type component; the within-class component can shorten them	
Negative lag excludes a dominant slow driver	Alternation and a slow component can coexist; no mechanism allocation follows	
A rounded SE boundary is a safe inclusive margin	The decision is strict; the rescaling boundary is exclusive and does not validate dependence	
Frozen M2 forecasts are 4.6% too long	The best level shift reduces the untruncated clock scale by 4.57%; that is not a remaining-wait bias	
The gap after a long outburst is about five days longer	A 14% clock-scale ratio; days need an explicitly chosen reference scale	

The original exploratory numbers are retained. These corrections do not alter the frozen primary comparison, its outcomes or the predeclared class-rule result.

Novelty wording narrowed in A10l. The draft framing called this the first preregistered held-out forecast test of dwarf-nova recurrence. The refreshed search found a held-out point-forecast study of V363 Lyr by Grigorievskiy and colleagues, in the 2014 ITISE proceedings (pp. 520–531), with a single test outburst. The claim now specifies preregistered, probabilistic and duration-based.

Draft corrections in A10m. Semantic review corrected an overstatement of complete flare recovery, distinguished an unseen-waveform qualification failure from unperformed X4 recovery, defined the standardised timing predictors, and gave the within-class likelihood statistic its proper class-model reference. The article's assumptions table now states calculated results and current limits rather than attributing information exclusively to class or physical disc state. The uniform-end lower bound is rounded directly from the full-precision evidence, and the SE failure boundary is explicitly approximate when printed. Reproduction commands preserve committed evidence and state their retained-input dependencies. The primary comparison is unchanged.

S5. Reproduction and navigation

From the repository root, without new archive requests:

These public-only checks use the committed aggregates and figure assets:

```
python3 manuscript/figures.py --check
make -C manuscript check
make -C manuscript
```

Recomputing scores and exploratory fits requires the retained inputs outside Git. The saved-score checker needs the completed evaluation run. The three subsequent analyses also need the retained 1910–1959 development inputs. Outputs below go to scratch paths; committed evidence is preserved.

```
python3 scripts/check_a10g_evaluation.py \
  --directory outputs/a10h_live_repair_5e39aeb \
  --output outputs/a10m_reproduction/score_audit.json
uv run --frozen --extra analysis \
  python scripts/a10j_postresult_review.py \
  --output outputs/a10m_reproduction/a10j_summary.json
uv run --frozen --extra analysis \
```

```
  python scripts/a10k_exploratory_audit.py
uv run --frozen --extra analysis \
  python scripts/a10k_secondary_audit.py
```

The retained run includes raw archive replies, observer identities and per-issuance scores. Its archive SHA-256 and manifest are in `data/provenance/a10i_evaluation_1960_2017/archive.json`. The committed aggregates support the ledger values, but do not replace these inputs for a full recomputation. The published protocol is at https://doi.org/10.5281/zenodo.24053338.

Navigation:

- `manuscript/claims.csv`: every claim, with its status, evidence, commit, reproduction command and limits.
- `manuscript/estimands.md`: allowed and forbidden readings.
- `coordination/decisions.md`: D013–D030.
- `docs/audits/`: every review, including A10i (the result), A10j (Claude's review), A10k (Codex's audit) and A10l (sources and drafts).
- `docs/outcome-access-ledger.md`: who saw which outcomes, and when.
- `manuscript/text/for-agents.md`: the agent guide.

References

Cannizzo, John K., and Janet A. Mattei. 1992. "On the Long-Term Behavior of SS Cygni." *The Astrophysical Journal* 401: 642. https://doi.org/10.1086/172092.

Price, A., A. A. Henden, G. Foster, V. Petriew, R. Huziak, R. James, M. D. Koppelman, et al. 2007. "SS Cygni Outburst Predictors and Long Term Quasi-Periodic Behavior." *Publications of the Astronomical Society of the Pacific* 119 (862): 1361–66. https://doi.org/10.1086/524413.